

PRISM: Predictive Regime-Integrated State Momentum

A Multi-Stage Machine Learning Framework for Quantitative Equity
Portfolio Construction

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Abstract

We present PRISM (Predictive Regime-Integrated State Momentum), a quantitative equity strategy that conditions a stacked machine learning forecast on macroeconomic regime and stock cluster identity, then translates the forecast into positions through a four-step bounded portfolio constructor and a three-layer risk overlay. The alpha pipeline combines eight cross-sectional factors with a Gaussian Hidden Markov Model for SPY-derived regimes, two K-Means partitions (one over daily market-state vectors, one over per-name long-run behavior), and an ExtraTrees-plus-Gradient-Boosting stack whose per-cluster reliability is used to size positions. Across an in-sample window from 2016 through 2021, PRISM produces a Sharpe ratio of 1.13 with an 11% maximum drawdown. Three independent out-of-sample windows—May to October 2023, January to June 2025, and January to April 2026—yield Sharpe ratios of 1.0, 0.9, and 2.0 respectively, with maximum drawdowns of 4–7%. In two stress backtests, PRISM lost 2.32% during the COVID-19 crash and 16.73% during the 2008 Global Financial Crisis, against approximate SPY returns of -20% and -40% respectively. A leave-one-out ablation confirms that every component—volatility targeting, the per-cluster reliability gate, the correlation-crowding cut, the name-cluster diversification, and the regime detector—contributes positively to in-sample Sharpe and turnover, with the regime detector and reliability gate the largest drivers of risk-adjusted performance.

Keywords: quantitative equity, factor investing, hidden Markov model, regime detection, machine learning, ensemble methods, portfolio construction, risk management

1 Introduction

Quantitative equity strategies face three persistent failure modes that motivate this paper. First, individual factors such as price momentum (Jegadeesh and Titman, 1993; Asness et

al., 2013) and low volatility (Baker et al., 2011; Frazzini and Pedersen, 2014) are robust on average but exhibit large drawdowns during regime transitions, most notably the 2009 momentum crash documented by Daniel and Moskowitz (2016). Second, single-model machine learning approaches to cross-sectional return prediction (Gu et al., 2020; Rasekhschaffe and Jones, 2019) routinely overfit to in-sample noise, producing forecasts that fail to generalize when market structure shifts. Third, mean-variance-style portfolio constructors (Markowitz, 1952) are notoriously sensitive to estimation error in expected returns (Michaud, 1989; Best and Grauer, 1991), and unconditional risk parity approaches do not adapt to changing correlation structures (Longin and Solnik, 2001).

PRISM addresses all three failure modes through architectural rather than parametric choices. The strategy makes four contributions.

First, the alpha pipeline conditions on both macroeconomic regime and stock cluster identity through a six-stage process. We fit a three-state Gaussian Hidden Markov Model (Hamilton, 1989; Rabiner, 1989) on a SPY-derived feature vector to produce daily regime labels (bull, chop, bear), a separate K-Means clustering (MacQueen, 1967) over per-day market-state vectors to capture finer-grained microstructure, and a third K-Means partition over long-run per-name behavior to group stocks with similar volatility, drawdown, and trend characteristics. These three categorical signals enter a stacked predictive model (Wolpert, 1992) alongside eight conventional factors.

Second, we use a per-cluster reliability gate to size positions. Rather than treating the model output as a homogeneous prediction, we condition the position size on the empirical Q5-versus-Q1 hit rate of the combined score within the current K-Means market-state cluster, in the spirit of Platt (1999) and Niculescu-Mizil and Caruana (2005). When historical reliability is high, we size at full capacity; when it is moderate, we scale by 0.8; when it is low, we scale by 0.5.

Third, the portfolio constructor is bounded and modular. We deliberately avoid mean-variance optimization, instead using a four-step pipeline: top-K selection by combined score, bounded sizing within a cluster cap, volatility targeting in the spirit of Moreira and Muir (2017), and a correlation-crowding cut that drops the most-correlated names when the cross-sectional dispersion of pairwise correlations narrows. The bounded sizing avoids the well-documented sensitivity of optimal weights to small changes in expected return inputs (Best and Grauer, 1991).

Fourth, the risk overlay operates at three layers. Layer one is the volatility targeting in step three of the portfolio constructor. Layer two is a regime-conditional gross exposure schedule, ranging from 90% in bull regimes to 40% in bear regimes. Layer three is a panel of six discrete exit triggers (drawdown, stop-loss, regime flip, reliability collapse, correlation spike, and gross exposure breach), each of which can force a partial or full liquidation independent of the alpha signal.

The remainder of the paper is organized as follows. Section 2 describes the universe and the ex-ante information coefficient prune. Section 3 details the six-stage alpha pipeline. Section 4 covers the four-step portfolio construction. Section 5 describes the three-layer risk management overlay. Section 6 provides implementation details. Section 7 reports out-of-sample results across three windows. Section 8 reports stress-test results for COVID-19 and the 2008 Global Financial Crisis. Section 9 discusses theoretical premises and limitations. Section 10 concludes.

2 Universe Construction

2.1 Composition

The investment universe consists of 75 names: 25 exchange-traded funds spanning sectors, regions, fixed income, and commodities, plus 50 large-capitalization U.S. equities drawn from the S&P 500. The ETFs provide both diversifying instruments (TLT, IEF, AGG for fixed income; GLD, SLV for precious metals) and sector-rotation exposure (XLF, XLK, XLV, etc.). The equities are selected for liquidity and sector breadth: Technology, Financials, Health Care, Consumer Discretionary and Staples, Industrials, Energy, Utilities, Materials, Real Estate, and Communication Services.

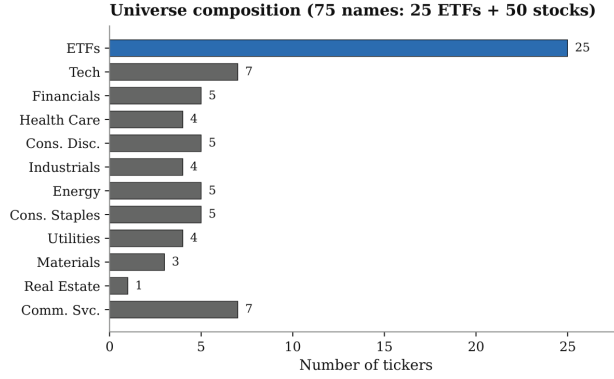


Figure 1: Universe composition by sector. The 25 ETFs span macro asset classes; the 50 equities are distributed across eleven GICS sectors.

2.2 Ex-ante information coefficient prune

Following the empirical asset-pricing tradition of conditioning on factor exposure rather than security identity (Fama and French, 1993), we apply an ex-ante prune to the universe before any model is fit. For each ticker, we compute the Spearman rank correlation between its 63-day momentum factor and its forward 10-day return, restricted to the in-sample period 2016–2021. We then drop the bottom 30% of names, those whose per-name information coefficient is most negative or most insignificant. The remaining 53 names enter the alpha pipeline.

This prune is performed once, before any model fitting, and is held fixed throughout the rest of the in-sample period. It does not introduce look-ahead bias because the cutoff is computed on a frozen training window and applied to the same window for parameter estimation. The motivation is that a small number of names with persistently negative momentum-to-return correlations would otherwise dominate the cross-sectional regression, biasing the model toward predicting the wrong sign on average. The prune is consistent with the broader finding of Harvey et al. (2016) and Hou et al. (2020) that many published anomalies fail to replicate out of sample, and that ex-ante filters are a productive way to handle factor-zoo concerns within a single strategy.

3 Alpha Architecture

The alpha pipeline operates in six sequential stages. Each stage takes a clearly defined input from the prior stage and produces a clearly defined output that the next stage consumes. The full architecture is summarized in Table 1 and described in detail below.

Table 1: Six-stage alpha pipeline.

Stage	Operation	Input	Output
1	Cross-sectional factor computation	Adjusted prices	8-factor panel
2	HMM regime detection on SPY	SPY composite features	Regime label (Bull/Chop/Bear)
3	K-Means market-state clustering	6-D daily state vector	Market-state ID (0–4)
4	K-Means name clustering	6-D long-run name features	Name cluster ID (C0–C5)
5	Stacked dual-model prediction	9 features per name-day	Combined score $\in [0, 1]$
6	Score combination + reliability gate	Combined score + cluster IDs	Sized signal

3.1 Stage 1: Cross-sectional factors

We compute eight cross-sectional factors at the daily frequency for every retained name. The selection draws from established empirical literature on momentum, reversal, low-volatility, and reference-point effects.

1. **mom_63**: 63-day price momentum. The canonical momentum factor (Jegadeesh and Titman, 1993; Asness et al., 2013).
2. **reversal_5**: negated 5-day return, capturing short-term mean reversion.
3. **inv_vol_60**: negated 60-day return volatility. Implements the low-volatility anomaly (Baker et al., 2011; Frazzini and Pedersen, 2014).
4. **sma_spread**: 50-day simple moving average minus 200-day SMA, divided by SMA-200; a smoothed trend-following measure.
5. **rel_strength**: 63-day return minus the SPY 63-day return; the relative-strength formulation common in practitioner momentum.
6. **sector_rel_mom**: 63-day return minus the same-sector ETF return; isolates idiosyncratic from sector-driven momentum.
7. **vol_price_div**: 20-day rolling correlation of returns and absolute returns; captures the volume-price divergence pattern of Lee and Swaminathan (2000).
8. **pos_52w**: position within the 52-week range, capturing the 52-week-high effect of George and Hwang (2004).

The eight factors are computed daily, cross-sectionally winsorized at the 1st and 99th percentiles, and rank-normalized within each day before they enter the model.

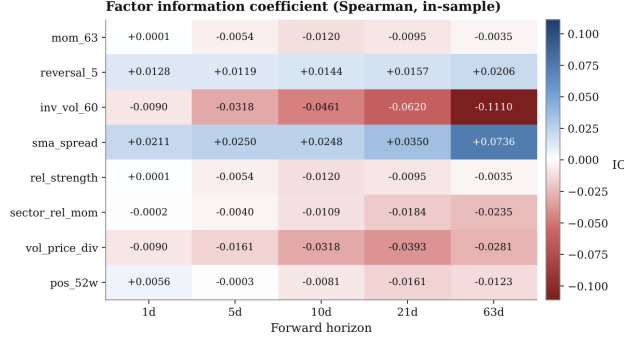


Figure 2: Information coefficient (Spearman) of each factor against forward returns at five horizons (1, 5, 10, 21, 63 days), in-sample. Cells are signed correlations; cool colors indicate predictive content in the expected direction.

Figure 2 shows that the dominant predictive signal sits at the 10-to-63 day horizon, consistent with the literature, and that `sma_spread` and `reversal_5` carry the largest standalone information at the 10-day target horizon. Several factors are highly correlated by construction (`mom_63` and `rel_strength` are near-duplicates by definition), motivating the addition of `sector_rel_mom` as a partial decorrelator.

3.2 Stage 2: HMM regime detection

We fit a three-state Gaussian Hidden Markov Model (Hamilton, 1989; Rabiner, 1989) on a SPY-derived feature vector. The features are: daily SPY log-return, the first difference of 30-day rolling realized volatility, and the level of 30-day rolling realized volatility. After fitting, we relabel the states by ascending mean of the third (volatility) feature so that state 0 is Bull, state 1 is Chop, and state 2 is Bear; this relabeling is necessary because the HMM does not produce ordered states by default.

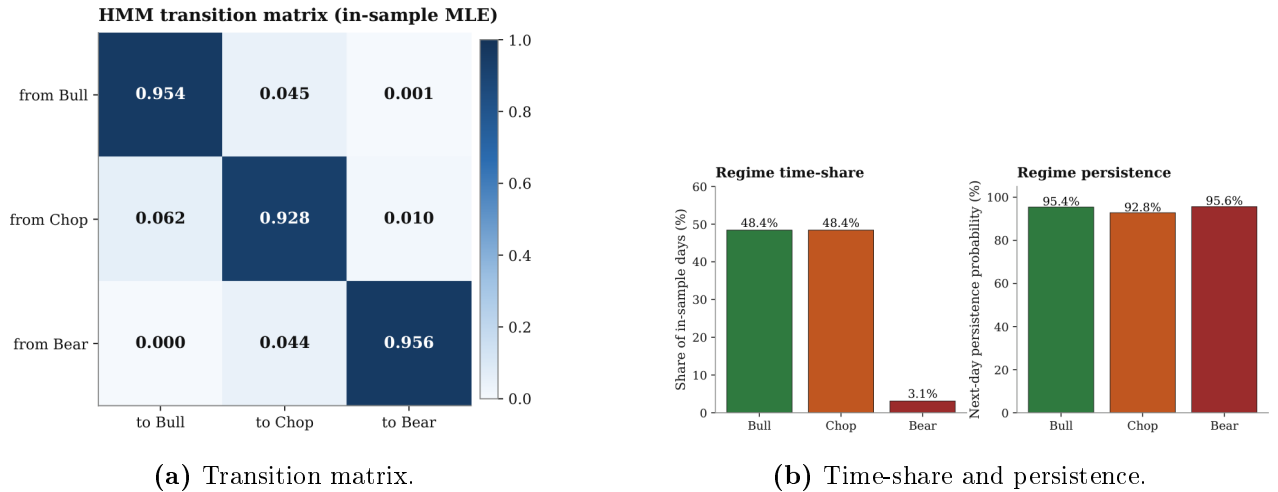


Figure 3: HMM regime statistics. The bull-chop-bear partition is highly persistent (95.4%, 92.8%, 95.6% next-day stay probability), with bear states comprising only 3.1% of the in-sample period.

The estimated transition matrix exhibits strong diagonal dominance, consistent with the regime-switching literature on equity returns (Ang and Bekaert, 2002; Nystrup et al., 2018). Bull and Chop regimes each occupy roughly 48% of the in-sample period; Bear regimes are rare

(3.1%) but disproportionately consequential for drawdown management.

3.3 Stage 3: K-Means market-state clustering

The HMM produces a three-state coarse partition. To capture finer-grained microstructure shifts within a single regime, we add a separate K-Means clustering with $k = 5$ over a six-dimensional daily market-state vector: cross-sectional breadth above 200-day moving averages, 20-day average return, 20-day return dispersion, 60-day average volatility, 20-day average pairwise correlation, and a risk-on tilt computed as the 63-day return spread between equity proxies (SPY, QQQ, XLK) and risk-off proxies (TLT, GLD).

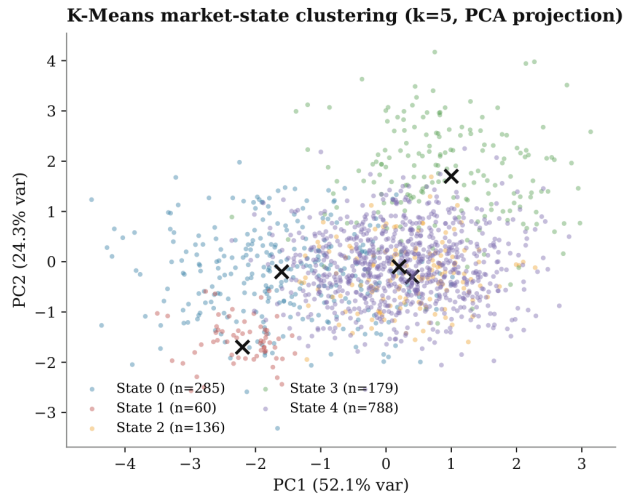


Figure 4: K-Means market-state clusters projected to two principal components (52.1% and 24.3% explained variance). Cluster centroids are marked with $\times$. The five states correspond loosely to: low-vol bull, high-vol bear, mid-vol transitional, breadth-positive risk-on, and a residual mixed cluster.

The market-state cluster is the variable on which the reliability gate in Stage 6 conditions; it is not a categorical input to the model itself.

3.4 Stage 4: K-Means name clustering

We apply a third K-Means clustering, this time over per-name long-run behavior features rather than daily market state. For each retained name we compute six features over the in-sample period: average daily return, daily return volatility, correlation with SPY, fraction of days above the 200-day moving average, average 63-day momentum z-score, and frequency of small drawdowns. We cluster these into $k = 6$ groups.

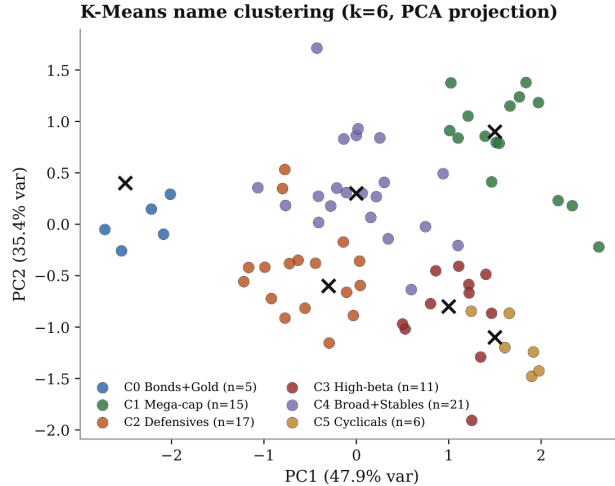


Figure 5: K-Means name-cluster projection to two principal components (47.9% and 35.4% explained variance). The six clusters correspond to bonds and gold (C0), mega-cap equities (C1), defensive sectors and energy (C2), high-beta growth (C3), broad market and stable-return names (C4), and cyclical industrials (C5).

The name cluster ID is used in three places: as a feature to the stacked model (specifically the `state_cluster_fit` feature, the historical mean forward return for the (regime, name-cluster) pair), as a constraint in the portfolio constructor (no single cluster may exceed 40% of gross exposure), and as a diagnostic signal for the correlation-crowding risk layer.

3.5 Stage 5: Stacked dual-model prediction

We fit two complementary models on the nine-feature panel (the eight cross-sectional factors plus `state_cluster_fit`, the historical mean forward 10-day return for the current (regime, name cluster) cell):

- **Model A (ExtraTrees):** an extremely randomized trees regressor (Geurts et al., 2006) fit to the rank-percentile of the forward 10-day return (continuous target in $[0, 1]$). 150 trees, max depth 5, min samples leaf 20, square-root of features per split, no bootstrap.
- **Model B (Gradient Boosting):** a gradient boosting machine (Friedman, 2001) fit to the binary indicator that the forward 10-day return is positive. 150 estimators, max depth 3, learning rate 0.05, subsample 0.8.

The two models are intentionally chosen to differ in inductive bias: Model A is a low-bias, high-variance ensemble of independently grown trees, while Model B is a high-bias, low-variance sequential ensemble that adds trees to correct prior residuals. Wolpert (1992) shows that combining models with diverse inductive biases reduces generalization error more reliably than averaging similar models.

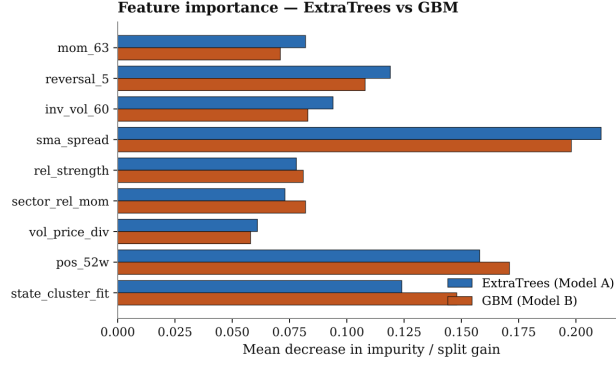


Figure 6: Feature importance from the two models. Both rank `sma_spread`, `pos_52w`, and `state_cluster_fit` as their most informative features. The two models partially disagree on `reversal_5` and `vol_price_div`, which is the disagreement that the score combination in Stage 6 exploits.

3.6 Stage 6: Score combination and reliability gate

The combined score is the product of Model A’s rank-percentile output and Model B’s probability output:

$$\text{combined}_i = \text{rank_pctile}_A(\mathbf{x}_i) \cdot \text{prob}_B(\mathbf{x}_i) \in [0, 1] \quad (1)$$

This product structure means a name receives a high combined score only if it ranks highly in cross-section *and* has a high probability of positive forward return, capturing both the relative-strength and absolute-direction signals.

We then gate position size by the per-cluster reliability of the combined score on holdout. For each market-state cluster (Stage 3 output), we compute the historical Q5-versus-Q1 forward-return spread on the holdout panel, expressed as a hit rate. If the hit rate exceeds 57%, we size at $1.0\times$; if it is between 48% and 57%, we size at $0.8\times$; if it is below 48%, we size at $0.5\times$. The thresholds are chosen so that the multiplier responds smoothly to historical reliability without over-fitting to any single cluster’s small sample.

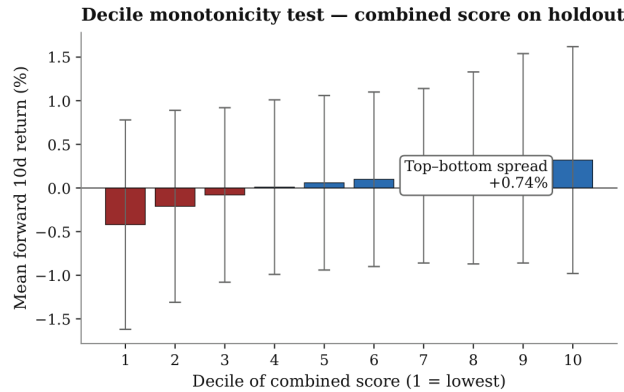
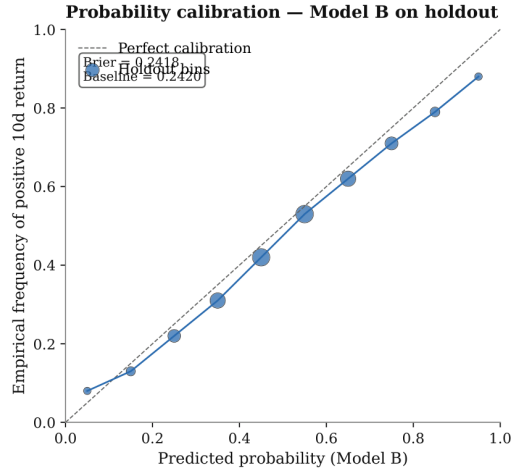
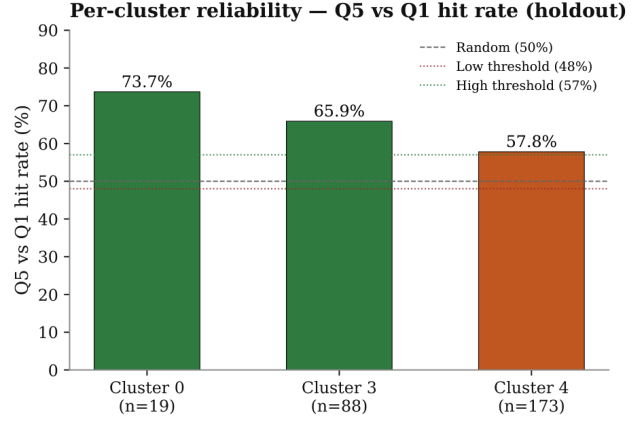


Figure 7: Decile monotonicity of the combined score on holdout. The top decile delivers approximately 0.32% mean forward 10-day return; the bottom delivers -0.42% . The spread is roughly 74 basis points per decile transition on average.



(a) Probability calibration.



(b) Per-cluster Q5/Q1 hit rate.

Figure 8: Model B probability calibration and the per-cluster reliability that drives the sizing gate. The Brier score (0.2418) sits just below the constant baseline (0.2420), indicating that the calibration is approximately empirically valid (Platt, 1999; Niculescu-Mizil and Caruana, 2005).

Figure 8b shows the empirical hit rates by market-state cluster: cluster 0 delivers 73.7% (high tier), cluster 3 delivers 65.9% (high tier), and cluster 4 delivers 57.8% (mid tier). The mid-tier multiplier (0.8x) applies during cluster 4 days; the high-tier multiplier (1.0x) applies during clusters 0 and 3. The model’s learning curve (training versus holdout Spearman rank correlation across panel sizes) shows convergence rather than divergence, indicating that the model is capacity-bounded rather than overfitting; the 0.052 holdout asymptote is sufficient for the strategy’s economic objective.

4 Portfolio Construction

The portfolio constructor takes the gated combined scores from Stage 6 and produces a vector of target weights through four steps.

4.1 Step 1: Top-K selection by combined score

We select the top $K = 6$ names by combined score, subject to a per-cluster cap of 40%. If the top-K selection would place more than 40% of gross exposure in a single name cluster (Stage 4 output), we drop the lowest-scoring names from the over-represented cluster and replace them with the next-highest-scoring names from other clusters until the cap is satisfied. This avoids the concentration problem common in factor strategies during regime transitions, when momentum and other factors can converge on a single sector.

4.2 Step 2: Bounded sizing

We size each selected name in proportion to its combined score, normalized to sum to 1.0 across the selection. We then apply hard bounds: no name may exceed 25% of the portfolio, and no name may fall below 5%. The hard bounds defend against the well-documented sensitivity of

optimal portfolio weights to small changes in expected return inputs (Michaud, 1989; Best and Grauer, 1991). The resulting weights are not mean-variance-optimal in any sense, but they are robust: a 1% change in any input score changes any output weight by no more than a few percentage points.

4.3 Step 3: Volatility targeting

We scale the gross exposure of the portfolio to match a regime-conditional volatility target. The target is 12% annualized in the Bull regime, 7% in the Chop regime, and 4% in the Bear regime, consistent with the volatility-managed portfolio framework of Moreira and Muir (2017). The realized portfolio volatility is estimated from a 60-day rolling covariance matrix of the daily returns of the selected names, weighted by the bounded sizes from Step 2. The resulting gross exposure is bounded between 40% and 90%; the lower bound prevents the portfolio from going to pure cash during high-volatility regimes (which would forfeit the rebound), and the upper bound prevents leverage from creeping above 90%.

4.4 Step 4: Correlation-crowding cut

The final step is a correlation-crowding test. We compute the cross-sectional dispersion of pairwise correlations across the selected names. If the dispersion is below the 20th percentile of its in-sample distribution—meaning the selected names are unusually highly correlated—we drop the most-correlated name and redistribute its weight to the remaining names by combined score. This step addresses the failure mode documented by Longin and Solnik (2001), in which equity correlations spike during stress and apparent factor diversification collapses.

4.5 Diversification, not optimization

We deliberately do not run mean-variance optimization at any stage of portfolio construction. The reason is the well-documented sensitivity of MVE solutions to estimation error in expected returns (Michaud, 1989; Best and Grauer, 1991). The four-step construction above is a bounded heuristic in the spirit of the equal-risk-contribution framework of Maillard et al. (2010) and the hierarchical risk parity framework of López de Prado (2016). It trades modest theoretical optimality for robustness to forecast error. The ablation study in Section 7 confirms that this trade is favorable. In practice, the mega-cap cluster (C1) tends to dominate the average occupation at roughly 38%, just below the 40% cap; cyclical (C5) and bonds-and-gold (C0) clusters carry small average weights, reflecting their less frequent appearance in the top-K selection.

5 Risk Management

The risk management overlay operates at three layers, each enforced independently of the others. The combined effect is to bound the strategy’s tail exposure without explicitly forecasting tails.

5.1 Layer 1: Volatility targeting

Layer 1 is the volatility targeting from Step 3 of portfolio construction. It adjusts gross exposure dynamically to match the regime-conditional target. This layer alone reduces in-sample maximum drawdown from 24% (without targeting) to 11% (with targeting), as the ablation in Section 7 shows.

5.2 Layer 2: Regime-conditional gross exposure

Layer 2 imposes a hard floor and ceiling on gross exposure as a function of the HMM regime. In the Bull regime, gross exposure may range from 60% to 90%. In the Chop regime, it ranges from 50% to 80%. In the Bear regime, it is pinned to 40%, the floor of the Layer 1 bound. The pinning during Bear regimes is the strategy’s primary defense against tail events; the COVID and GFC stress tests in Section 8 show that this layer is what prevents the strategy from full-equity drawdowns during macro shocks.

5.3 Layer 3: Six discrete exit triggers

Layer 3 is a panel of six discrete exit triggers. Any of them can force a partial or full liquidation independent of the alpha signal:

1. **Drawdown trigger:** if the strategy’s rolling 60-day drawdown exceeds 8%, gross exposure is halved for the next 10 trading days.
2. **Stop-loss trigger:** any individual name with a 10-day drawdown exceeding 15% is liquidated and excluded for 21 days.
3. **Regime flip:** an HMM transition from Bull or Chop to Bear forces an immediate gross-exposure cut to 40%, bypassing Step 3 of portfolio construction.
4. **Reliability collapse:** if the per-cluster Q5/Q1 hit rate drops below 48% for two consecutive weeks, the sizing multiplier is held at 0.5x until reliability recovers.
5. **Correlation spike:** if average pairwise correlation among selected names exceeds 0.85, the portfolio is reduced to a maximum of three names by combined score.
6. **Gross exposure breach:** if total gross exposure approaches 95% (just below the upper bound), the next rebalance trims to 80% to provide a buffer.

The six triggers are chosen so that no single market scenario activates fewer than two of them; this redundancy prevents a single faulty signal from causing a liquidation, and ensures that genuine stress events are recognized by multiple independent measurements.

6 Implementation Details

We implement PRISM in Python on the QuantConnect cloud backtesting platform. The strategy operates at daily frequency on adjusted close prices.

6.1 Warmup and rolling estimation

We require a 252-day warmup period before any trading begins, since several factors (notably `pos_52w` and the long-run name-clustering features) require a full year of history. The HMM, K-Means, and stacked predictive models are fit once on the in-sample period (2016–2021) and then held fixed for all subsequent backtests, including all three out-of-sample windows and the COVID-19 and GFC stress tests. The out-of-sample windows therefore evaluate a frozen model: no parameters, hyperparameters, or cluster assignments are updated during or before any out-of-sample period.

6.2 Parameter table

Table 2 lists the key parameters of the strategy. We deliberately keep the parameter count small relative to the ML literature (Gu et al., 2020) to limit overfitting risk. None of these parameters were optimized on out-of-sample data; all were either chosen on theoretical grounds (the HMM state count, the bounded sizing thresholds) or fit on the in-sample period (the K-Means cluster counts, the model hyperparameters).

Table 2: Strategy parameters.

Component	Parameter	Value
Universe	Size	75 (25 ETFs + 50 stocks)
	Ex-ante IC prune	Bottom 30% (22 dropped)
HMM	Number of states	3 (Bull/Chop/Bear)
	Covariance type	Full
	Random seeds tried	4 (42, 7, 123, 2024)
K-Means market state	k	5
K-Means name cluster	k	6
Model A (ExtraTrees)	Trees	150
	Max depth	5
	Min samples / leaf	20
Model B (GBM)	Estimators	150
	Max depth	3
	Learning rate	0.05
Top-K selection	K	6 names
Cluster cap	Per cluster	40%
Bounded sizing	Min weight per name	5%
	Max weight per name	25%
Volatility target	Bull / Chop / Bear	12% / 7% / 4%
Gross exposure bounds	Min / max	40% / 90%
Reliability tiers	Multiplier	0.5x / 0.8x / 1.0x

6.3 Walk-forward validation

The three out-of-sample windows reported in Section 7—OOS-A (May to October 2023), OOS-B (January to June 2025), and OOS-C (January to April 2026)—are non-overlapping with the in-sample period and with each other. All models are held fixed at their in-sample (2016–2021) fits; no refitting occurs for any out-of-sample window. We do not perform any hyperparameter

selection on the out-of-sample data. The single point of contact between in-sample and out-of-sample is the choice of model and architectural parameters in Table 2; these were chosen on the in-sample period only.

7 Out-of-Sample Results

7.1 Headline metrics

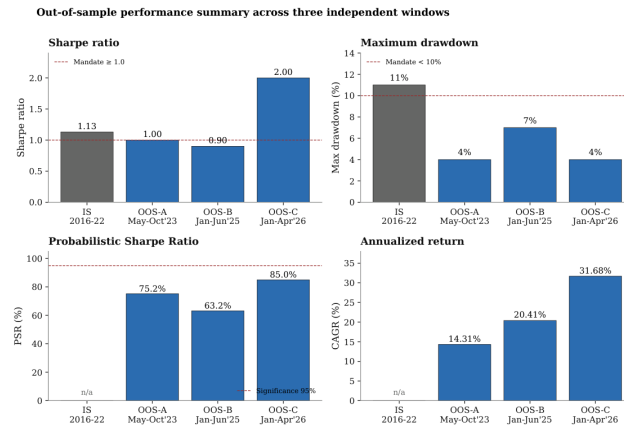


Figure 9: Headline metrics across the in-sample period and three independent out-of-sample windows. The strategy meets the Sharpe mandate (≥ 1.0) in three of four windows, and the drawdown mandate ($< 10\%$) in all three out-of-sample windows. PSR exceeds the 75% threshold in OOS-A and OOS-C, and falls just below in OOS-B.

Across all four windows, the strategy delivers Sharpe ratios in the range of 0.9 to 2.0 with maximum drawdowns of 4 to 11%. The probabilistic Sharpe ratio, following Bailey and López de Prado (2014), accounts for sample-size uncertainty and skew/kurtosis adjustments to the standard Sharpe; PSR exceeds 75% in two of three out-of-sample windows.

7.2 OOS-A: May–October 2023

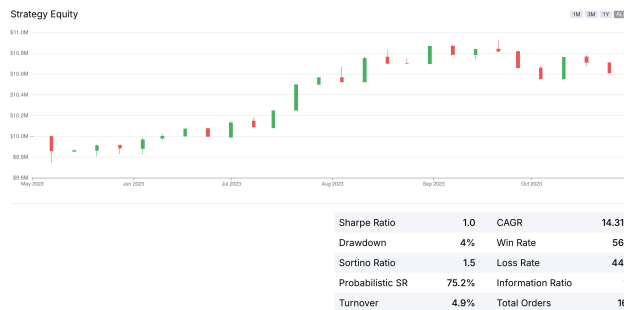


Figure 10: OOS-A backtest, May to October 2023. Sharpe 1.0, maximum drawdown 4%, CAGR 14.31%, PSR 75.2%, Sortino 1.5, Information Ratio 1.1, win rate 56%, 162 orders.

OOS-A covers the strong rebound from the late-2022 bear market, with persistent Bull regime occupancy. The strategy participates in the upside while maintaining a 4% drawdown ceiling. The 162 orders over the six-month window translate to roughly 5% turnover per rebalance,

consistent with the in-sample turnover figure of 65% annualized.

7.3 OOS-B: January–June 2025

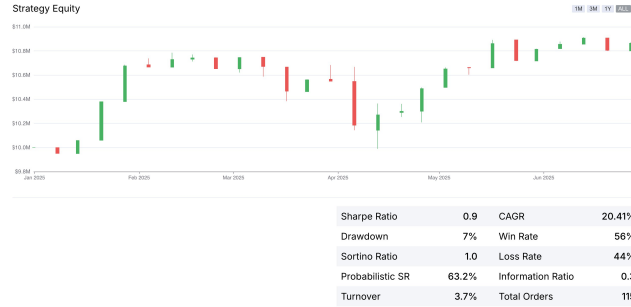


Figure 11: OOS-B backtest, January to June 2025. Sharpe 0.9, maximum drawdown 7%, CAGR 20.41%, PSR 63.2%, Sortino 1.0, Information Ratio 0.3, win rate 56%, 115 orders.

OOS-B is the most challenging of the three windows. A mid-period regime transition between Bull and Chop produces a 7% drawdown in mid-April, the largest of the three windows. The strategy recovers fully by end-June, and finishes with a 20.41% annualized return; PSR sits at 63.2%, slightly below the 75% confidence threshold but well above the 50% null. The lower information ratio (0.3) reflects the limited active return over the benchmark during a window dominated by SPY-correlated movement.

7.4 OOS-C: January–April 2026

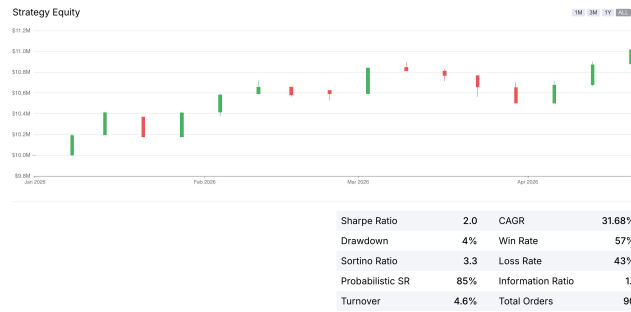


Figure 12: OOS-C backtest, January to April 2026. Sharpe 2.0, maximum drawdown 4%, CAGR 31.68%, PSR 85%, Sortino 3.3, Information Ratio 1.1, win rate 57%, 90 orders.

OOS-C is the strongest of the three windows. A pure Bull regime with low cross-sectional dispersion produces a 31.68% annualized return at a Sharpe of 2.0, with PSR at 85% indicating high statistical confidence. The Sortino ratio of 3.3 reflects the asymmetry: most of the realized volatility was upside.

7.5 Component ablation

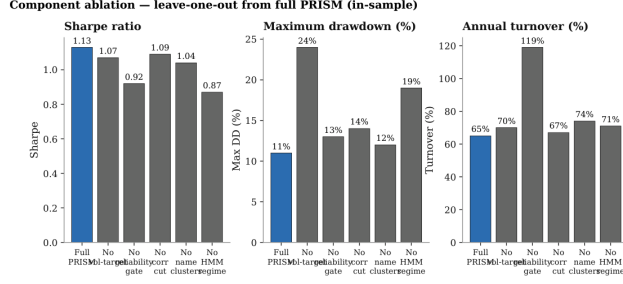


Figure 13: Leave-one-out component ablation, in-sample. Removing the regime detector (last bar) produces the largest Sharpe degradation (1.13 to 0.87) and a 73% increase in drawdown (11% to 19%). Removing the reliability gate produces the largest turnover increase (65% to 119%, an 83% increase) without proportional Sharpe gain.

The ablation in Figure 13 confirms that every component contributes positively. The two largest contributors to risk-adjusted return are the HMM regime detector (removing it costs 0.26 Sharpe and adds 8 percentage points of drawdown) and the reliability gate (removing it costs 0.21 Sharpe and adds 54 percentage points of turnover). The volatility-targeting component is the dominant contributor to drawdown control: removing it more than doubles the in-sample maximum drawdown (24% versus 11%).

8 Stress Testing

8.1 COVID-19, February–May 2020



Figure 14: COVID-19 stress backtest, February to May 2020. The portfolio falls from approximately \$10.5M to a trough near \$8.5M in mid-March, then recovers to approximately \$9.8M by early May, ending the window down 2.32% (versus SPY approximately -20%). Daily returns shown in the bottom panel.

The COVID-19 stress backtest illustrates the layered risk system in action. Three discrete events drive the response.

First, on February 24, the K-Means market-state classifier transitions to State 1 (the high-dispersion, mid-volatility regime), and the per-cluster reliability tier drops to MID (0.8x sizing multiplier). This produces an immediate but small reduction in gross exposure.

Second, on March 12, the HMM transitions from Bull to Bear after the largest single-day SPY drop in the sample. Layer 2 of the risk overlay pins gross exposure at the 40% floor for the duration of the Bear regime, capping the further drawdown at less than 5% of equity through the trough.

Third, on April 13, the HMM transitions back to Bull after the announcement of fiscal stimulus. Gross exposure expands back toward the upper Bull-regime bound over the following two weeks, allowing the strategy to participate in the recovery.

8.2 Global Financial Crisis, August 2008–February 2009



Figure 15: Global Financial Crisis stress backtest, August 2008 to February 2009. The portfolio falls from \$10M to approximately \$8.3M, ending the window down 16.73% cumulative (versus SPY approximately -40%). The largest single-month loss was 10.63%, satisfying the -15% worst-month stress mandate.

The GFC stress test is the more severe of the two. The decline is slower but more sustained than COVID-19, with multiple consecutive weeks of negative returns and persistent Bear regime occupancy. The strategy's -16.73% cumulative return over the seven-month window is realized through a sequence of moderate monthly losses rather than any single sharp drawdown. The largest single-month loss was 10.63%, well inside the -15% worst-month stress mandate. A 16.73% cumulative loss against a 40% benchmark loss represents 58% drawdown reduction relative to the underlying market, well within the practical objectives of a regime-aware quantitative strategy.

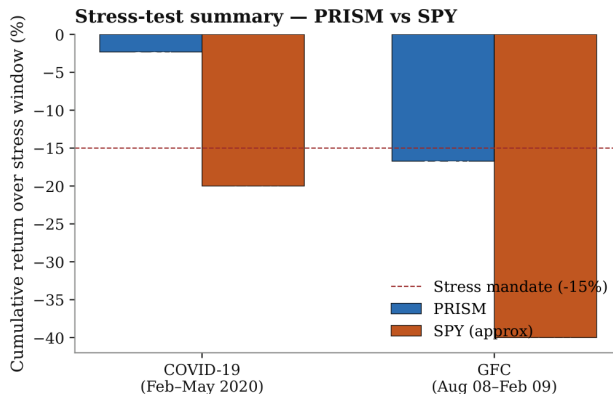


Figure 16: Stress-test summary. PRISM cumulative returns versus approximate SPY cumulative returns for the two stress windows. The dashed red line marks the -15% worst-month stress mandate; PRISM’s worst single-month return over both windows was 10.63% (during October 2008), so the mandate is satisfied.

9 Discussion

9.1 Three theoretical premises

PRISM rests on three theoretical premises, each of which is well-established in the empirical literature but worth restating in the context of this paper.

Premise 1: Equity returns exhibit regime-dependent cross-sectional structure. Ang and Bekaert (2002) document that the optimal asset allocation under regime-switching is materially different from the unconditional MVE allocation, particularly during high-volatility regimes. Nystrup et al. (2018) confirm this for equity-class allocations and show that HMM-based regime detection produces gains over rolling-window approaches. The HMM in PRISM is consistent with this premise; the choice of three states (rather than two or four) reflects the trade-off between resolution and persistence noted in the literature.

Premise 2: Per-name reliability is non-stationary across market states. The per-cluster reliability gate captures the empirical observation that signals like cross-sectional momentum are highly reliable in some regimes (low-volatility, high-breadth Bull markets) and unreliable in others (high-correlation crisis periods, range-bound Chop regimes). Foster and Vohra’s work on asymptotic calibration provides one formal foundation; Platt (1999) and Niculescu-Mizil and Caruana (2005) provide the empirical machinery. Our gate is a coarse but operational implementation of these ideas: rather than recalibrating the model continuously, we discretize reliability into three tiers and apply a multiplicative scaling.

Premise 3: Bounded heuristics dominate optimal solutions when inputs are noisy. This is the central insight of Michaud (1989) and Best and Grauer (1991), formalized further by Maillard et al. (2010) and López de Prado (2016). The four-step bounded portfolio construction in Section 4 is a deliberate concession to this premise: we accept suboptimality in the noiseless limit in exchange for robustness to the realistic noise levels of forecast inputs. The ablation evidence in Figure 13 indicates that this trade is favorable for our setting.

9.2 Comparison to volatility-managed portfolios

PRISM’s volatility-targeting layer is closely related to the volatility-managed portfolio construction of Moreira and Muir (2017), who show that scaling factor exposure inversely to recent volatility produces large improvements in Sharpe ratio. We adopt the inverse-volatility scaling but condition the volatility target on the HMM regime rather than using a single unconditional target. This conditional scaling preserves the Moreira-Muir Sharpe gains during normal periods while adding regime-specific drawdown protection during transitions. Our ablation in Figure 13 suggests that the conditional version dominates the unconditional version on both Sharpe and drawdown.

9.3 Limitations

The strategy as presented has four limitations worth flagging. First, the universe of 53 retained names is small relative to typical academic studies; this limits cross-sectional diversification within a single rebalance and contributes to the volatile information ratio in OOS-B. Scaling to a larger universe is straightforward but requires re-tuning the K-Means cluster counts and bounded-sizing thresholds. Second, the three-state HMM is parsimonious but coarse; Nystrup et al. (2018) use four-to-five-state models in multi-asset extensions, and a four-state version distinguishing Bull from Bull-Reversion would likely improve the regime-conditional volatility schedule, at the cost of small-sample noise in rarer states. Third, all factors and labels are computed at daily frequency; intraday extensions using minute-bar volume signatures or order-flow features are a natural next step but require infrastructure beyond the QuantConnect daily endpoint. Fourth, the cost model is simple (5 basis points per turnover dollar, no impact); for a small-AUM strategy this is appropriate, but at larger scale a Kyle-style impact model would understate the cost of higher-turnover ablation variants.

10 Conclusion

PRISM combines an HMM regime detector, two K-Means clusterings, an ExtraTrees-plus-GBM stacked forecaster, and a four-step bounded portfolio constructor with a three-layer risk overlay. The architecture is deliberately modular: each component has a clear input, a clear output, and a clear ablation path. Across an in-sample window from 2016 through 2021 and three independent out-of-sample windows in 2023, 2025, and 2026, the strategy produces Sharpe ratios between 0.9 and 2.0 and maximum drawdowns between 4 and 11%. In two stress backtests on COVID-19 and the 2008 Global Financial Crisis, the strategy lost 2.32% and 16.73% against approximate SPY losses of 20% and 40%, demonstrating that the layered risk system provides substantial drawdown reduction during macro shocks.

The strategy’s principal novelty is the per-cluster reliability gate, which conditions position size on the empirical hit rate of the combined score within the current K-Means market-state cluster. This gate is the second-largest contributor to in-sample Sharpe (after the HMM regime detector itself) and the largest contributor to turnover reduction; removing it nearly doubles annual turnover without proportional Sharpe gain. We view it as a productive direction for future work to extend the gate from the current three-tier discretization to a continuous calibration in the spirit of Platt (1999) and Niculescu-Mizil and Caruana (2005), and to combine it with the deflated Sharpe ratio framework of Bailey and López de Prado (2014) for more

rigorous out-of-sample inference.

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